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Studierichting: Informatica
Oefeningenreeks 4A nr. 35.1

$$\begin{aligned}\int 2^x dx &= \int e^{x \ln 2} dx \\ t = x \ln 2 \left(\Rightarrow dt = \ln 2 dx \Rightarrow dx = \frac{dt}{\ln 2} \right) \\ \int e^t \frac{dt}{\ln 2} \\ &= \frac{1}{\ln 2} \int e^t dt \\ &= \frac{1}{\ln 2} e^t + C \\ x \ln 2 &= t \\ &= \frac{1}{\ln 2} e^{x \ln 2} + C \\ &= \frac{1}{\ln 2} 2^x + C\end{aligned}$$

References